

YUXUAN NAI

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EDUCATION

ZJU-UIUC Institute

B.E in Mechanical Engineering

Zhejiang University, China

B.E in Mechanical Engineering

University of Illinois at Urbana-Champaign, U.S.

B.E in Mechanical Engineering

Zhejiang University, China

PhD in Control Science and Engineering

Champaign, Illinois / Zhejiang, China

09/2022 – Present

09/2022 – 09/2024 & 09/2025 – 05/ 2026

GPA: 3.97/4.3

09/2024 – 09/ 2025

GPA: 3.92/4.0

09/2026(Soon)

RESEARCH INTERESTS

Deep Reinforcement Learning, Tactile Based Manipulation, Legged Locomotion

PUBLICATIONS AND WORKING PAPER

* Indicates corresponding authors

- Beyond Robustness: Learning Unknown Dynamic Load Adaptation for Quadruped Locomotion on Rough Terrain
Leixin Chang, **Yuxuan Nai**, Hua Chen*, Liangjing Yang*
Accepted at *IEEE International Conference on Robotics and Automation (ICRA)*, 2025. [\[Project\]](#) [\[Paper\]](#)
- Bilevel Optimization of Heli-Logging Assignment for a Quadrotor with Cable-Suspended Payload
Yuxuan Nai, Ding Jiang, Ziyin Han, Chengyu Yang, Sheng Cheng*, Naira Hovakimyan*
Under Review at *IEEE Robotics and Automation Letters (RAL)*, 2025.

RESEARCH

UMI Motion Generation Humanoid Loco-Manipulation

Physical Intelligence Lab, ZJU-UIUC

Advisor: Prof. Hua Chen

03/2026 – 06/2026

- Use UMI data to generate full-body motion sequences for G1 from wrist-mounted RGB images alone.
- Use Flowmatching to integrate motion prior information, enabling motion generation zero-shot.
- Use WBC, and incorporate ResMimic as an additional module to improve hand precision.

UMI Guided Loco-Manipulation

Physical Intelligence Lab, ZJU-UIUC

Advisor: Prof. Hua Chen

03/2026 – 06/2026

- Built a two-layer loco-manipulation framework transferring manipulation skills to a bipedal platform.
- Trained a DiT Diffusion Policy on UMI data to predict 6D end-effector trajectories from RGB.
- Trained an RL whole-body controller in Isaac Lab to track SE(3) commands on the bipedal robot.

Learning Based Stable Pose Selection with Tactile Modality

RoboTouch Lab, UIUC

Advisor: Prof. Wenzhen Yuan

05/2025 – 09/2025

- Realized stable pose selection via a tactile-feedback-driven learning-based planner.
- Enabled stability prediction of poses by training a model with data collected on the Taccel simulator.

Bilevel Optimization of Quadrotor Payload System

AVIATE Center, UIUC

Advisor: Dr. Sheng Cheng (Postdoctoral Fellow), Prof. Naira Hovakimyan

10/2024 – 10/2025

- Developed minimum-snap trajectory with flatness for smooth transitions between taut and slack cable states.
- Improved trajectory fidelity to real maneuvers via bilevel optimization of release parameters.
- Validated control strategies through simulation testing on Gazebo.

RL for Legged Unknown Dynamic Load Adaptation

Physical Intelligence Lab, ZJU-UIUC

Advisor: Prof. Hua Chen, Prof. Liangjing Yang

05/2024 – 07/2024

- Facilitated blind walking by implementing a teacher-student reinforcement learning framework.
- Enabled quadruped locomotion under unknown payloads with a proprioception-based payload estimator.
- Validated through MuJoCo simulations and sim-to-real deployment.

Unmanned Mining Truck Route Planning

Academic Training SRTP, ZJU

Advisor: Prof. Jian Chen

05/2023 – 05/2024

- Simplified route planning for unmanned mining trucks, overcoming traditional method limitations.
- Developed a two-step strategy: front-end path planning and back-end trajectory optimization.
- Used Hard-Constraints and Minimum-Snap to ensure safe, dynamically feasible paths.

INTENSHP

A3 Humanoid Perceptive Parkour

AGIBOT

Advisor: Taiyu Li

01/2026 – 03/2026

- Developed a unified policy for full-scale A3 humanoid locomotion.
- Trained a single policy for multiple walking-base locomotion tasks.
- Implemented perceptive parkour for terrain-aware agile locomotion.

PROJECT

Three-axis robotic Arm Control for Trajectory Following

UIUC

Course Project from Robot Dynamics and Control (ME446, A⁺)

01/2025 – 05/2025

- Designed trajectories to accomplish multiple tasks—egg pressing, obstacle avoidance, and peg-in-hole insertion.
- Developed a PID controller with a feedforward term for enhanced performance.
- Validated the approach on a CRS robotic arm, achieving smooth and rapid task execution.

Identification of Informative COVID-19 English Tweets

UIUC

Course Project from Programming Methods for Machine Learning (ECE364, A⁺)

01/2025 – 05/2025

- Design a classification network to identify informative tweets.
- Explored and evaluated various training methods for the network.
- Achieved 0.89 accuracy, ranking 3rd overall in the competition.

Reaction-Based Motion Planning with Deep Reinforcement Learning

ZJU-UIUC

Project Reproduction

02/2024 – 03/2024

- Reproduced a project integrating Artificial Potential Field (APF) with DDPG to mitigate local minima issues.
- Developed advanced motion planning leveraging DDPG’s continuous actions.
- Validated approach in ROS Gazebo simulations, achieving robust navigation with dynamic obstacles.

HONORS AND AWARDS

China National Scholarship

10/2025

First Prize, Zhejiang University Scholarship

10/2025

Second Prize, Zhejiang University Scholarship

10/2024

Gold Medal, Award on Zhejiang International College Student Innovation Competition

07/2024

Silver Medal, Award on Challenge Cup Contest of Zhejiang Province

05/2024

First Prize, Award on The 16th Dandelion Entrepreneurship Contest of Zhejiang University

03/2024

Third Prize, Award on Concrete Canoe Design Competition of Zhejiang University

06/2023

Second Prize, Award on Undergraduate Structural Design Competition of Zhejiang University

05/2023

LEADERSHIP

SRTP Program Leader

05/2023 – 05/2024, ZJU

- Led an undergraduate research team focused on autonomous path planning for unmanned vehicles.

Summer Research Program Leader

05/2023 – 08/2023, ZJU

- Led an undergraduate team researching path planning integrated with reinforcement learning.

Photography Club Leader

08/2023 – 08/2024, ZJU-UIUC

- Found the first photography club at ZJU-UIUC and ran the club as a leader.

ACADEMIC COLLABORATORS

- ZJU-UIUC: Prof. Liangjing Yang, Prof. Hua Chen
- UIUC: Prof. Naira Hovakimyan, Prof. Wenzhen Yuan, Dr. Sheng Cheng (Postdoctoral Fellow)